THE INTERNAL HOM OF PRESHEAVES OF MODULES COMMUTES WITH LIMITS AND THE INTERNAL HOM OF SHEAVES DOES TOO

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites no labelled external item (the adjointness used is discussed in the same section), so no navigation stubs were needed.

Allowed standard prerequisites (editorial).

- Restriction to the localized category is both a left and a right adjoint. Stacks Project, Modules on Sites, section on internal hom (adjointness of j_U^*). Gives preservation of limits by restriction, which the sectionwise computation needs.

Lemma 0.1. *Internal hom and (co)limits. Let $\mathcal{C}$ be a category and let $\mathcal{O}$ be a presheaf of rings.*

(1) *For any presheaf of $\mathcal{O}$-modules $\mathcal{F}$ the functor*
$$PMod(\mathcal{O}) \longrightarrow PMod(\mathcal{O}), \quad \mathcal{G} \longmapsto \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$$
commutes with arbitrary limits.

(2) *For any presheaf of $\mathcal{O}$-modules $\mathcal{G}$ the functor*
$$PMod(\mathcal{O}) \longrightarrow PMod(\mathcal{O})^{opp}, \quad \mathcal{F} \longmapsto \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$$
commutes with arbitrary colimits, in a formula
$$\mathcal{H}om_{\mathcal{O}}(\operatorname{colim}_i \mathcal{F}_i, \mathcal{G}) = \lim_i \mathcal{H}om_{\mathcal{O}}(\mathcal{F}_i, \mathcal{G}).$$

Suppose that $\mathcal{C}$ is a site, and $\mathcal{O}$ is a sheaf of rings.

(3) *For any sheaf of $\mathcal{O}$-modules $\mathcal{F}$ the functor*
$$Mod(\mathcal{O}) \longrightarrow Mod(\mathcal{O}), \quad \mathcal{G} \longmapsto \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$$
commutes with arbitrary limits.

(4) *For any sheaf of $\mathcal{O}$-modules $\mathcal{G}$ the functor*
$$Mod(\mathcal{O}) \longrightarrow Mod(\mathcal{O})^{opp}, \quad \mathcal{F} \longmapsto \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$$
commutes with arbitrary colimits, in a formula
$$\mathcal{H}om_{\mathcal{O}}(\operatorname{colim}_i \mathcal{F}_i, \mathcal{G}) = \lim_i \mathcal{H}om_{\mathcal{O}}(\mathcal{F}_i, \mathcal{G}).$$

Proof. Let $\mathcal{I} \to PMod(\mathcal{O})$, $i \mapsto \mathcal{G}_i$ be a diagram. Let U be an object of the category $\mathcal{C}$. As j_U^* is both a left and a right adjoint we see that $\lim_i j_U^* \mathcal{G}_i = j_U^* \lim_i \mathcal{G}_i$. Hence we have
$$\begin{aligned} \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \lim_i \mathcal{G}_i)(U) &= \operatorname{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \lim_i \mathcal{G}_i|_U) \\ &= \lim_i \operatorname{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}_i|_U) \\ &= \lim_i \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G}_i)(U) \end{aligned}$$

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by definition of a limit. This proves (1). Part (2) is proved in exactly the same way. Part (3) follows from (1) because the limit of a diagram of sheaves is the same as the limit in the category of presheaves. Finally, (4) follow because, in the formula we have

$$\mathrm{Mor}_{Mod(\mathcal{O})}(\mathrm{colim}_i \mathcal{F}_i, \mathcal{G}) = \mathrm{Mor}_{PMod(\mathcal{O})}(\mathrm{colim}_i^{PSh} \mathcal{F}_i, \mathcal{G})$$

as the colimit $\mathrm{colim}_i \mathcal{F}_i$ is the sheafification of the colimit $\mathrm{colim}_i^{PSh} \mathcal{F}_i$ in $PMod(\mathcal{O})$. Hence (4) follows from (2) (by the remark on limits above again). $\square$